

Geometric Intuition ● Beginner

Zooming back out to the picture, now that the numbers behind it are no longer mysterious.

Data points → Candidate line → Residuals → Total squared error → Best-fitting line

Every candidate line produces its own set of residuals and its own SSE. OLS is the search over all possible lines for the one specific line with the smallest possible SSE.

WHY THE LINE DOESN'T HIT EVERY POINT

Regression finds the line that provides the best *overall* fit according to the chosen loss function (squared error) — not a line forced through every point. Forcing the line through every point (like connecting dots) would overfit to noise and generalize terribly to new data.

Optimization Intuition ● Intermediate

Reframing the same problem as "minimize a cost function" — the language that unlocks gradient descent, neural networks, and beyond.

COST / LOSS FUNCTION

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

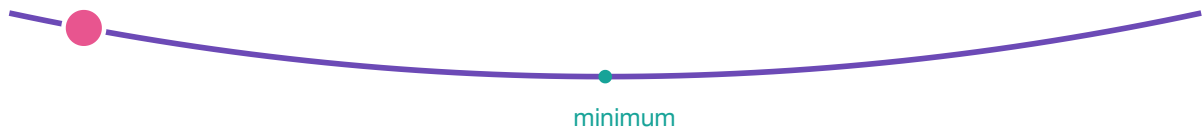
MSE is just SSE divided by the number of observations — the *average* squared error instead of the total.

Metric	What it is
SSE	Total squared error (sensitive to how much data you have)
MSE	Average squared error (comparable across dataset sizes)
RMSE	Square root of MSE (back in the original units of y)

Gradient Descent — the visual version first

Mountain / bowl-shaped cost surface → Start somewhere → Calculate direction of steepest descent → Take a step → Repeat → Reach minimum

Interactive — Gradient Descent



Position = 1.00 | Cost = 1.980

The curve is a simplified cost surface. A tiny learning rate converges slowly; too large a learning rate can overshoot and bounce.

Term	Meaning
Learning rate	How big a step is taken at each iteration
Gradient	The direction (and steepness) of the slope at the current point
Iteration	One step of "measure gradient, move, repeat"
Local minimum	A low point the algorithm might settle at — for plain linear regression's bowl-shaped cost surface, there's only one minimum, so this isn't a concern here
Convergence	The point where further steps barely change the cost — the algorithm has effectively arrived

OLS vs. Gradient Descent

OLS	Gradient Descent
Analytical, closed-form solution	Iterative optimization
Very efficient for small/medium feature sets	Useful for large-scale problems (millions of rows/features)
Directly calculates coefficients	Updates coefficients repeatedly until convergence

In practice, `scikit-learn`'s `LinearRegression` commonly uses least-squares solvers rather than requiring you to manually implement gradient descent — but understanding gradient descent matters because it generalizes to models where no closed-form solution exists (like most of deep learning).